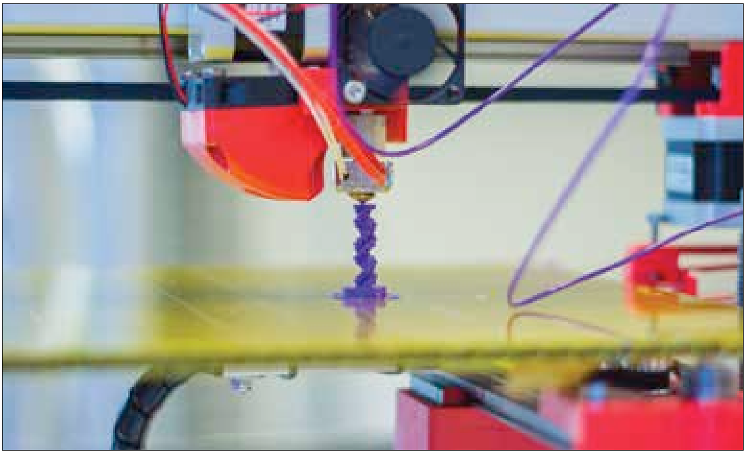

3D printer printing an object

are powered by tiny yet powerful low-power chips called microprocessors. They handle all the functions of the devices they live in.

In your smartphone, microprocessors encode your voice and talks to the GSM/CDMA hardware which connects to your mobile company to complete your call. It also handles encryption while simultaneously monitoring battery level during the call. For 3G video services such as video calling, microprocessors encode the camera video output to enable transmission of video at lower bandwidth or lower cost (based on your preferences). All the snazzy graphics that we see on mobile phone screens are handled by microprocessors.

You probably interacted with it a minute ago...

The touchscreens on smartphones send data to the microcontroller that calculates the position where the touch occurred and takes appropriate decisions (by modifying the elements on the screen) to carry out your desired tasks. It is the microcontroller that detects when your phone is tilted and modifies the elements on screen to adjust to the new orientation and enables gaming by tilting the device instead of using keys. The keypad on feature phones sends the key press data depending on which a change in menu occurs on the screen and the next possible option is shown to the user. All the apps we use, the webpages we visit are drawn (rendered) on screen by the microcontroller.